Topic 4: Evaluation Frameworks (RAGAS and TruLens)

Introduction

As Large Language Models (LLMs) become increasingly integrated into real-world applications, ensuring the quality, accuracy, and reliability of their responses has become a critical requirement. Although modern LLMs such as Gemini and GPT generate highly fluent responses, they may still produce hallucinations, retrieve irrelevant information, or generate incomplete answers. Therefore, an effective evaluation mechanism is required to assess the performance of LLM-based applications.

Evaluation frameworks provide automated methods to measure the quality of AI-generated responses. They help developers determine whether the generated answers are factually correct, relevant to the user's query, grounded in the retrieved context, and free from hallucinations.

For the proposed internship project, two complementary evaluation frameworks are selected:

- **RAGAS (Retrieval-Augmented Generation Assessment)** – evaluates the quality of the RAG pipeline.
- **TruLens** – monitors, traces, and evaluates the complete LLM application.

Using both frameworks together provides a comprehensive evaluation strategy that measures both response quality and overall system performance.

What is an Evaluation Framework?

An evaluation framework is a software library that automatically measures the quality and performance of AI-generated responses using predefined metrics.

Instead of manually checking thousands of responses, developers can use evaluation frameworks to generate objective quality scores and identify weaknesses in the system.

An evaluation framework answers questions such as:

- Is the answer factually correct?
- Is the response relevant to the user's question?
- Is the answer supported by retrieved documents?
- Did the retriever select the correct context?
- Is the model hallucinating?
- How fast is the system responding?
- What is the overall quality of the application?

Why Evaluation Frameworks are Required

Modern AI systems consist of multiple components, including document retrieval, embedding generation, vector search, LLM inference, and response generation.

An error in any of these stages may produce an incorrect final answer.

For example:

User Question:

"What is Machine Learning?"

Possible Problems:

- Wrong document retrieved
- Missing document retrieval
- Hallucinated answer
- Irrelevant response
- Slow response generation

Evaluation frameworks automatically detect these issues and provide measurable quality scores.

RAGAS (Retrieval-Augmented Generation Assessment)

Introduction

RAGAS is an open-source evaluation framework specifically designed for Retrieval-Augmented Generation (RAG) systems.

Unlike traditional evaluation metrics such as BLEU or ROUGE, RAGAS evaluates both the retrieval process and the generated response.

Its primary objective is to determine whether the retrieved documents support the generated answer and whether the retrieval pipeline is functioning effectively.

What does RAGAS do?

RAGAS evaluates the complete RAG workflow by analyzing:

- User Question
- Retrieved Context
- Generated Response

It calculates multiple evaluation metrics that measure retrieval quality, answer quality, and hallucination detection.

RAGAS Evaluation Metrics

1. Faithfulness

Faithfulness measures whether the generated answer is supported by the retrieved documents.

Example:

Retrieved Document:

"Python is an interpreted programming language."

Generated Answer:

"Python is an interpreted programming language."

Faithfulness Score: High

Generated Answer:

"Python was developed by Microsoft."

Faithfulness Score: Low

This metric is one of the most important indicators for detecting hallucinations.

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2. Answer Relevancy

Measures whether the generated answer directly addresses the user's question.

Question:

"What is Artificial Intelligence?"

Good Response:

"Artificial Intelligence enables computers to perform tasks requiring human intelligence."

Poor Response:

"Artificial Intelligence is becoming popular worldwide."

Although related, the second response does not answer the question completely.

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3. Context Precision

Measures how many retrieved documents are actually useful.

Example:

Retrieved Documents:

- Machine Learning Basics
- Neural Networks
- Deep Learning

Precision is high.

Retrieved Documents:

- Machine Learning
- Football Rules
- Cooking Recipes

Precision becomes low because irrelevant documents were retrieved.

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4. Context Recall

Measures whether the retriever successfully found all important information.

Example:

Question:

"Explain DBMS."

Expected Topics:

- Database
- Tables
- SQL
- Transactions
- Normalization

If only "Database" and "Tables" are retrieved, recall is low because important information is missing.

Advantages of RAGAS

- Designed specifically for RAG systems.
- Automatically detects hallucinations.
- Measures retrieval quality.
- Measures answer quality.
- Easy integration with LangChain.
- Open-source and widely adopted.
- Supports automated benchmarking.

TruLens

Introduction

TruLens is an open-source evaluation and monitoring framework designed for complete LLM applications.

While RAGAS focuses primarily on retrieval quality and answer correctness, TruLens provides end-to-end monitoring of the AI pipeline.

It records every interaction and generates detailed evaluation reports.

What does TruLens do?

TruLens continuously monitors the complete AI application by recording:

- User query
- Prompt
- Retrieved context
- Generated response
- Execution time
- Token usage
- API cost

- Evaluation scores
- Feedback history

Developers can later inspect these logs to identify system issues and optimize performance.

TruLens Evaluation Metrics

1. Groundedness

Measures whether the generated response is supported by the retrieved context.

This metric is conceptually similar to the Faithfulness metric in RAGAS.

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2. Context Relevance

Measures whether the retrieved documents are relevant to the user's query.

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3. Answer Relevance

Evaluates whether the generated response satisfies the user's request.

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4. Toxicity

Detects harmful, offensive, or inappropriate language.

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5. Sentiment Analysis

Determines whether the generated response has a positive, neutral, or negative sentiment.

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6. Custom Feedback Functions

Developers can define organization-specific evaluation rules.

Examples include:

- Medical compliance
- Financial regulations
- Company policies
- Educational standards

Advantages of TruLens

- End-to-end monitoring.
- Interactive dashboard.
- Tracks response latency.
- Tracks API cost.
- Stores prompts and responses.
- Supports custom evaluation metrics.
- Production-ready.
- Easy debugging and performance analysis.

Why Use Both RAGAS and TruLens Together?

Although both frameworks evaluate AI systems, they serve different purposes.

RAGAS specializes in evaluating the quality of the RAG pipeline, whereas TruLens focuses on monitoring and analyzing the complete AI application.

Using both frameworks provides a comprehensive evaluation strategy.

RAGAS is responsible for:

- Faithfulness
- Context Precision
- Context Recall
- Answer Relevancy
- Hallucination Detection

TruLens is responsible for:

- Application Monitoring
- Prompt Tracking
- Response Logging
- Latency Monitoring
- Token Usage
- API Cost Analysis
- Feedback Dashboard
- Custom Evaluation Metrics

Together, they provide complete visibility into both AI quality and system performance.

Integration of RAGAS and TruLens in the Proposed Project

The proposed internship project integrates both evaluation frameworks into a single RAG pipeline.

System Workflow

```
                         User
                          │
                          ▼
                   User Question
                          │
                          ▼
               LangChain RAG Pipeline
                          │
            ┌─────────────┴──────────────┐
            ▼                            ▼
 Document Retrieval                Query Embedding
            │
            ▼
       FAISS Vector DB
            │
            ▼
  Retrieved Document Chunks
            │
            ▼
       Gemini LLM
            │
            ▼
     Generated Response
            │
       ┌────┴─────────────────────────────┐
       ▼                                  ▼
   RAGAS                              TruLens
       │                                  │
 Faithfulness                    Prompt Logging
 Context Precision               Response Logging
 Context Recall                  Latency Tracking
 Answer Relevancy                Cost Monitoring
 Hallucination Detection         Dashboard & Feedback
       │                                  │
       └───────────────┬──────────────────┘
                       ▼
         Final Evaluation Report
```

```
                 |
                 ▼
        Response to the User
```

Comparison of RAGAS and TruLens

Feature	RAGAS	TruLens
Primary Purpose	Evaluate RAG quality	Monitor complete LLM application
Hallucination Detection	Yes	Yes
Faithfulness Evaluation	Yes	Through Groundedness
Context Precision	Yes	No
Context Recall	Yes	No
Answer Relevancy	Yes	Yes
Response Logging	No	Yes
Dashboard	No	Yes
Latency Monitoring	No	Yes
Cost Monitoring	No	Yes
Custom Metrics	Limited	Extensive
Production Monitoring	Limited	Excellent

Proposed Technology Integration

Component	Technology
Large Language Model	Gemini API
RAG Framework	LangChain
Embedding Model	Sentence Transformers (all-MiniLM-L6-v2)
Vector Database	FAISS
Chunking Strategy	Recursive Character Text Splitter
Retrieval Evaluation	RAGAS

Component	Technology
Application Monitoring	TruLens
Backend	Python + Flask
Frontend	React.js

Conclusion

The proposed evaluation architecture integrates both RAGAS and TruLens to build a reliable, transparent, and high-quality Retrieval-Augmented Generation (RAG) system. RAGAS evaluates the correctness of document retrieval and generated responses using metrics such as Faithfulness, Context Precision, Context Recall, and Answer Relevancy. TruLens complements this evaluation by monitoring the complete AI application, including prompt execution, retrieved context, latency, token usage, API cost, and feedback analysis.

By combining these two frameworks, the proposed system not only detects hallucinations and improves response quality but also provides comprehensive monitoring and performance analysis. This integrated approach follows current industry best practices for developing trustworthy and production-ready AI applications.